

Data Encryption Standard (DES)

Source Code:

```
from Crypto.Cipher import DES
import base64

def pad(text):
    padding_length = 8 - (len(text) % 8)
    return text + chr(padding_length) * padding_length

def unpad(text):
    return text[:-ord(text[-1])]

def des_encrypt(plain_text, key):
    cipher = DES.new(key.encode(), DES.MODE_ECB)
    padded_text = pad(plain_text)
    encrypted_text = cipher.encrypt(padded_text.encode())
    return base64.b64encode(encrypted_text).decode()

def des_decrypt(encrypted_text, key):
    cipher = DES.new(key.encode(), DES.MODE_ECB)
    decrypted_text = cipher.decrypt(base64.b64decode(encrypted_text)).decode()
    return unpad(decrypted_text)

key = input("Enter an 8-character key: ")
plain_text = input("Enter the plain text: ")
encrypted_text = des_encrypt(plain_text, key)
print(f"Encrypted: {encrypted_text}")
decrypted_text = des_decrypt(encrypted_text, key)
print(f"Decrypted: {decrypted_text}")
```

Output/ Implementation:

```
DES.py x
DES.py > ...
1  from Crypto.Cipher import DES
2  import base64
3
4  def pad(text):
5      padding_length = 8 - (len(text) % 8)
6      return text + chr(padding_length) * padding_length
7
8  def unpad(text):
9      return text[:-ord(text[-1])]
10
11 def des_encrypt(plain_text, key):
12     cipher = DES.new(key.encode(), DES.MODE_ECB)
13     padded_text = pad(plain_text)
14     encrypted_text = cipher.encrypt(padded_text.encode())
15     return base64.b64encode(encrypted_text).decode()
16
17 def des_decrypt(encrypted_text, key):
18     cipher = DES.new(key.encode(), DES.MODE_ECB)
19     decrypted_text = cipher.decrypt(encrypted_text.encode())
20     return unpad(decrypted_text).decode()
21
22 if __name__ == '__main__':
23     key = input("Enter an 8-character key: ")
24     plain_text = input("Enter the plain text: ")
25     encrypted_text = des_encrypt(plain_text, key)
26     decrypted_text = des_decrypt(encrypted_text, key)
27     print("Encrypted: " + encrypted_text)
28     print("Decrypted: " + decrypted_text)
```

PROBLEMS 2 OUTPUT DEBUG CONSOLE TERMINAL PORTS SPELL CHECKER 2

PS C:\Users\thava\OneDrive\Desktop\Currently working\SC> & C:/Users/thava/AppData/Local/Programs/Python/Python38-64/Scripts/python C:/Users/thava/OneDrive/Desktop/Currently working/SC/DES.py
Enter an 8-character key: helloguy
Enter the plain text: Secured Computinh
Encrypted: SR+P8Lt1RVl6lC4uq08H2lAAud6ymtr8
Decrypted: Secured Computinh